A Size-Aware Graph Embedding Approach to Remote Sensing Image Captioning With Object Relative Size Information

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***Abstract*—Remote sensing image captioning is the task of automatically generating descriptive texts for remotely sensed scenes and objects. A common shortcoming of existing methods is the inadequate consideration of object size, which often leads to captions that either omit size information or provide imprecise size descriptions. To overcome this deficiency, we develop a novel framework composed of three modules: 1) an object confirmation and relative size estimation module; 2) a graph construction and graph convolution module; and 3) a caption generation module. Our framework comprehensively characterizes object size to generate more quantitatively informative captions. Furthermore, we introduce a new evaluation metric, SizeNum-Meteor, designed to explicitly evaluate the correctness of object count and relative size information in generated captions. This provides a more comprehensive assessment, as standard metrics typically neglect the evaluation of object size. In addition, we construct extended benchmarks by enriching existing datasets with explicit annotations of object count and relative size. Extensive experiments on three existing benchmark datasets (i.e., UC Merced (UCM), Sydney, and remote sensing image captioning dataset (RSICD)] and the benchmarks we construct (i.e., UCM-N-S, Sydney-N-S, and RSICD-N-S) demonstrate that our framework achieves superior performance on both standard metrics and the proposed SizeNum-Meteor.**

***Index Terms*—Image captioning, object size, remote sensing.**

I. Introduction

REMOTE sensing plays a pivotal role in modern geospatial analysis and environmental monitoring [1], [2], [3], [4], [5], [6], [7], [8]. Traditional research in this field has primarily focused on tasks such as object detection, land-use classification, and scene recognition, which aim to extract structured information from the intricate details of remote sensing imagery. In recent years, remote sensing image captioning has become a critical task with the rapid growth in demand for intelligent interpretation. However, most existing studies mainly emphasize semantic interpretation while lacking explicit modeling of quantitative attributes such as object size. In practical remote sensing applications such as urban planning, land-use monitoring, and environmental assessment, object size serves as a fundamental quantitative attribute for geo-observation and interpretation. Analysts often rely on the spatial extent of objects such as the area of farmlands, the coverage of construction zones, or the width of water bodies to evaluate land-use intensity, detect environmental changes, or assess urban expansion. In this context, we aim to integrate both semantic and quantitative understanding into remote sensing image captioning.

Remote sensing image captioning involves generating coherent and semantically accurate natural language descriptions that capture the visual content of ground scenes observed from aerial or satellite perspectives. The emergence of large language models (LLMs) has propelled this task into a new era, enabling more sophisticated and contextually rich interpretations. While numerous captioning methods have been developed for both natural and remote sensing images, significant challenges remain in achieving optimal accuracy, generalization, and domain-specific relevance.

Sections I-A–I-E will review the relevant literature, analyze current deficiencies in the field, itemize our novel contributions, itemize our novel contributions to advancing remote sensing image captioning, and present the outline of the article.

A. Literature Review of Natural Image Captioning

Natural image captioning aims to automatically generate textual captions for given images. Traditional methods often utilized template-based systems [9], which rely on predefined sentence structures to describe image contents.

With the progress of deep learning, natural image captioning has entered a new era. Vinyals et al. [10] introduced the Show-and-Tell model, an end-to-end method combining CNN-based visual encoders and LSTM-based language decoders. Han et al. [11] proposed an aerial image captioning method that integrates attention mechanisms to enhance the relevance and coherence of generated captions. Tang et al. [12] proposed a reinforcement learning-based method that leverages Transformer architectures to produce more fluent and context-aware captions. Recent efforts have explored lightweight and efficient captioning systems. Ramos et al. [13] developed SmallCap to reduce training complexity by conditioning caption generation

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on retrieved similar captions. Honda et al. [14] proposed a switching captioning model that incorporates reinforcement learning to improve the distinctiveness of generated captions.

Despite these advancements, these captioning methods often focus on general semantics while ignoring fine-grained information such as object count and size. Most generated captions use ambiguous terms like "some people" or "very big," which limits the usefulness of these captions in applications that require precise numerical or spatial interpretation.

B. Literature Review of Remote Sensing Image Captioning

Remote sensing image captioning adapts ordinary image captioning to the geospatial domain, where quantitative information regarding objects serves as crucial features for geo-observation.

In this area, a wide range of methods have been proposed. Wu et al. [15] proposed a dual-transformer architecture that employs Swin Transformer for multiscale visual feature extraction and a Transformer-based decoder (TrTr-CMR) for remote sensing image caption generation. Cheng et al. [16] proposed a multiscale feature extraction network (MST) equipped with a small-target attention module to effectively address the challenges associated with low resolution and small object recognition in remote sensing imagery. Meng et al. [17] developed a Mamba network (RSIC-GMamba), a hybrid method that integrates genetic algorithms and state–space modeling within the Mamba architecture to improve global context reasoning and robustness in remote sensing image captioning tasks. Das and Sharma [18] proposed a text-graph convolution network (GCN) combined with a multilayer LSTM encoder–decoder structure named "M-T-C," capturing semantic relationships between words at both sentence and corpus levels to enhance caption quality. Wang et al. [19] designed a multiview feature fusion and prompt method that integrates vision–language pretraining with a visual prefix mapping strategy to generate flexible captions. Das et al. [20] conducted a study based on an Efficient-Net framework, including decoding strategies such as greedy search and beam search, offering evaluations of encoder choice in remote sensing image captioning performance. Zhou et al. [21] proposed a FRIC method referred to as TR-FRIC to tackle the few-shot challenge by applying decoder-side optimization through ensemble and self-distillation techniques. Li et al. [22] proposed a refinement and rethinking attention mechanism that interacts with grid-level features and reevaluates multiple spatial regions during caption generation.

Despite these advancements, the methods of remote sensing image captioning still largely overlook quantitative information. However, quantitative information is of greater importance in remote sensing image captioning because Earth observation inherently demands precise and reliable captions, and the absence of such information limits the effectiveness of captions in practical geo-observation.

C. Deficiencies in Current Remote Sensing Image Captioning Research

To further illustrate the current status of remote sensing image captioning, Table I shows captions generated by several representative large models for a remote sensing image. The check marks (✓) and crosses (✗) indicate whether the models correctly identified the object count. In addition, the "N/A" entries in the size column highlight that existing models generally fail to provide size information. In contrast, our proposed framework (bottom row) not only gives the correct object count but also the relative size, marked as "Available" in red. Thus, even with such progress in existing works and models, remote sensing image captioning still exhibits three key deficiencies.

1) Existing remote sensing image captioning methods generally neglect the explicit object size descriptions. However, object size constitutes a crucial quantitative attribute in geo-observation, where accurate size descriptions are indispensable. The absence of explicit size descriptions often leads to captions that either omit size information altogether or provide vague and unreliable expressions, thereby reducing their practical value.

2) Existing evaluation metrics are not suitable for assessing the quality of generated captions from the aspect of object size. Conventional metrics such as Bleu, Meteor, and consensus-based image description evaluation (Cider) assess linguistic fluency and semantic similarity but fail to capture correctness of object size and count.

3) Remote sensing captioning currently lacks benchmark datasets annotated with object size information. Current remote sensing captioning datasets lack annotations of object size and count, making it difficult to train and evaluate models that aim to generate quantitatively informative captions.

D. Our Novel Contributions

To address the aforementioned deficiencies, we make three main contributions which are summarized as follows.

1) We develop a novel relative size-aware captioning framework. The framework explicitly models object relative size through an object confirmation and relative size estimation module, enabling the generation of captions that are more quantitatively precise and semantically informative. An example is given in the bottom row of Table I.

2) We design a new evaluation metric, SizeNum-Meteor. This metric explicitly assesses the correctness of object relative size and count information in generated captions, offering a more comprehensive evaluation than conventional metrics such as Bleu, Meteor, and Cider.

3) We introduce relative size-annotated remote sensing captioning datasets. By enriching existing datasets with object relative size and count annotations, we provide training and evaluation resources that facilitate the development of relative size-aware captioning framework.

E. Paper Outline

The article is organized as follows. Section II presents our remote sensing image captioning framework with object relative size, detailing object confirmation and size estimation, graph construction and graph convolution, and caption generation, followed by the overall objective and loss design. Section III presents the evaluation metric: we first review

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TABLE I
EXEMPLARY CAPTIONS GENERATED BY DIFFERENT MODELS FOR A REMOTE SENSING IMAGE

A remote sensing image	Model	Generated caption	Count	Size
	GPT-4o[23]	Two baseball fields are located in a green park, with one field in the upper-left and another in the lower-right corner	✓	N/A
	Claude 3.5[24]	An aerial image showing two orange-red baseball diamonds within a grassy area	✓	N/A
	Gemini 1.5 Pro[25]	A large open field with a single baseball field in the lower-right corner	✗	N/A
	LLaVA-NeXT[26]	The picture shows green grassland with two baseball fields visible	✓	N/A
	Vicuna-13B[27]	The image shows one baseball field in the middle of the park	✗	N/A
	InternLM-XComposer[28]	A park area with two baseball fields, surrounded by grass and trees	✓	N/A
	Ours	There are 2 baseball fields in the playground and their relative sizes range from 3496 to 4375	✓	Available

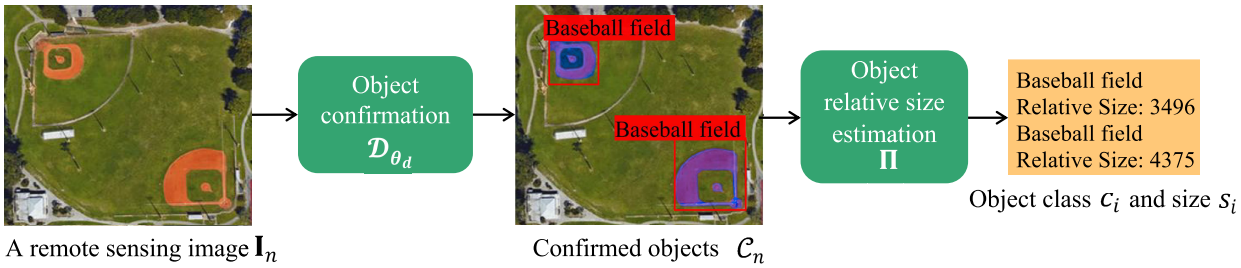


Fig. 1. Object confirmation and relative size estimation.

traditional Meteor and then propose SizeNum-Meteor tailored for quantitative correctness in captions. Section IV presents the datasets employed, including segmentation-labeled data, standard captioning benchmarks, and our extended sets with count and relative-size annotations. Section V presents experiments, training setup, metrics, and comparative results, alongside ablation analysis. Finally, Section VI concludes the article and discusses future directions.

II. NOVEL RELATIVE SIZE-AWARE CAPTIONING FRAMEWORK

We develop a novel framework that incorporates object relative size into remote sensing image captioning. Our proposed framework comprises three modules: object confirmation and relative size estimation, graph construction and graph convolution module, and caption generation. They will be presented one by one in the following subsections, followed by an overall summarized subsection.

A. Object Confirmation and Relative Size Estimation

In remote sensing imagery, object size is a critical quantitative attribute for geo-observation. While absolute size can be derived from pixel counts together with sensor calibration parameters such as resolution or altitude, in this work, we focus on relative size, defined as the number of pixels occupied by an object within an image. Relative size preserves proportional relationships among objects, and can be easily converted to absolute size when calibration data are available. We, therefore, adopt relative size as a practical proxy for object size in our framework.

In order to characterize object relative size, we develop an object confirmation and relative size estimation module, as illustrated in Fig. 1. It contains an object confirmation model and an object relative size estimation model Π.

A remote sensing image $\mathbf{I}_n$ is input into an object confirmation model $\mathcal{D}_{\theta_d}$ with parameters θ_d, which generates the segmented objects in mask (in purple in Fig. 1) with their predicted class labels

$$\mathcal{C}_n = \mathcal{D}_{\theta_d}(\mathbf{I}_n) \tag{1}$$

where $\mathcal{C}_n = \{c_{n,k}\}_{k=1}^{K}$ denotes the set of all predicted object class labels in image $\mathbf{I}_n$, $c_{n,k}$ is the class label of the kth object in image $\mathbf{I}_n$, and K is the total number of segmented objects with class labels in $\mathbf{I}_n$.

Each confirmed object is input into an object relative size estimation model Π, which generates the object relative size $s_{n,k}$ for the kth confirmed object.

Specifically, in the object confirmation model, we use the Mask R-CNN [29] object segmentation network to segment and classify objects within the remote sensing image.

In the object relative size estimation model Π, the Mask R-CNN network estimates the object relative size $s_{n,k}$ (i.e., the number of pixels occupied by the object) and normalizes the relative size within each image using min–max normalization

$$\hat{s}_{n,k} = \frac{s_{n,k} - \min(\mathcal{S}_n)}{\max(\mathcal{S}_n) - \min(\mathcal{S}_n)}, \quad \mathcal{S}_n = \{s_{n,k}\}_{k=1}^{K} \tag{2}$$

where $\mathcal{S}_n$ collects the unnormalized relative sizes of all confirmed objects in image $\mathbf{I}_n$, and $k \in \{1, \ldots, K\}$. The normalization is performed per image.

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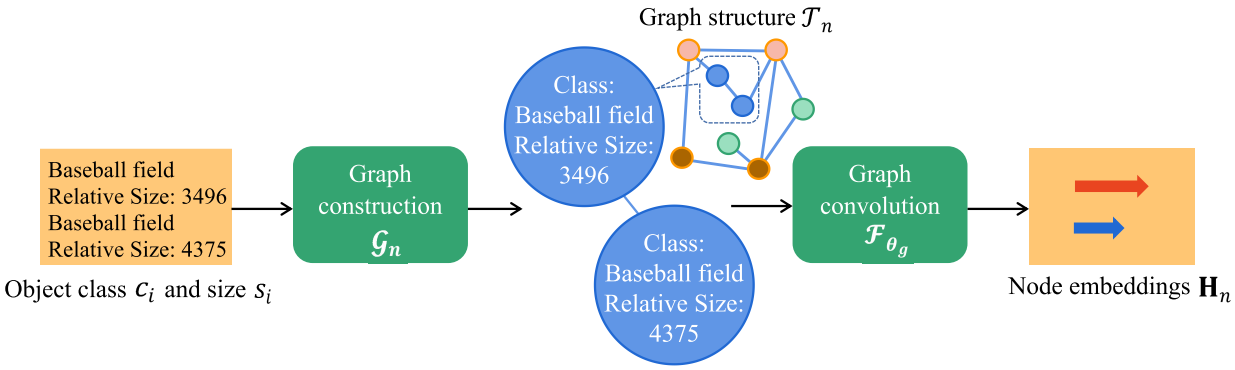


Fig. 2. Graph construction and graph convolution. The red and blue arrows denote node vectors produced from the two graph nodes after graph convolution model.

The kth segmented object is thus represented by a feature pair $(c_{n,k}, \hat{s}_{n,k})$, passed to the graph construction model $\mathcal{G}_n = (\mathcal{V}_n, \mathcal{E}_n)$.

B. Graph Construction and Graph Convolution

In order to model spatial relationships among objects (e.g., the "containment" relation in the remote sensing image $\mathbf{I}_n$, where confirmed objects such as baseball fields are contained within the playground), we develop a graph construction and graph convolution module, as illustrated in Fig. 2, where the lower part depicts a two-node structure corresponding to the remote sensing image $\mathbf{I}_n$. The "two-node structure" in Fig. 2 illustrates a simplified example in which two segmented objects (e.g., two baseball fields) are represented as graph nodes. The upper part presents the complete multinode graph that typically arises in large-scale remote sensing images. It contains the graph construction model $\mathcal{G}_n = (\mathcal{V}_n, \mathcal{E}_n)$ and graph convolution model $\mathcal{F}_{\theta_g}$ with parameters θ_g.

The object class c_i and relative size s_i generated from the object confirmation and relative size estimation module are input into the graph construction model $\mathcal{G}_n = (\mathcal{V}_n, \mathcal{E}_n)$ and the graph structure $\mathcal{T}_n = (\mathcal{V}_n, \mathcal{E}_n)$ is generated. The graph structure $\mathcal{T}_n$ is input into the graph convolution model $\mathcal{F}_{\theta_g}$ and the node embeddings $\mathbf{H}_n$ (which are represented by red and blue arrows in Fig. 2) are generated.

In the constructed graph structure $\mathcal{T}_n = (\mathcal{V}_n, \mathcal{E}_n)$, we use a graph-based data structure consisting of nodes $\mathcal{V}_n$ and edges $\mathcal{E}_n$. Each node $v_k \in \mathcal{V}_n$ represents a segmented object and is initialized with its semantic features, i.e., the feature vector $\mathbf{h}_{n,k}^{(0)} = [\,\phi(c_{n,k}), \hat{s}_{n,k}\,]$. where $\phi(c_{n,k})$ denotes the learned embedding of the object's class label, and $\hat{s}_{n,k}$ represents the normalized area ratio of the object within the image. Edges between nodes capture spatial relationships such as adjacency or containment. Edges $e_{k\ell} \in \mathcal{E}_n$ encode spatial relationships (e.g., containment) between objects, derived from their relative positions within the remote sensing image.

Specifically, each node $v_k \in \mathcal{V}_n$ is initialized with a feature vector $\mathbf{h}_{n,k}^{(0)}$

$$\mathbf{h}_{n,k}^{(0)} = \left[\,\phi\left(c_{n,k}\right)\;;\;\hat{s}_{n,k}\,\right] \in \mathbb{R}^d, \qquad \phi : \mathcal{C} \to \mathbb{R}^{d_1} \tag{3}$$

where $c_{n,k} \in \mathcal{C}$ is the class label of the kth object, $\phi(c_{n,k}) \in \mathbb{R}^{d_1}$ is its learnable class embedding (implemented via a lookup table $E \in \mathbb{R}^{|\mathcal{C}| \times d_1}$), $\hat{s}_{n,k} \in \mathbb{R}$ is the normalized relative size, and the operator $[\,\cdot\,;\,\cdot\,]$ denotes vector concatenation, thus the total dimension is $d = d_1 + 1$.

The adjacency matrix $\mathbf{A}_n \in \{0,1\}^{K \times K}$ encodes edge presence as

$$[\mathbf{A}_n]_{k\ell} = \begin{cases} 1, & \text{if a spatial relation exists between } v_k \text{and } v_\ell \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where $v_k, v_\ell \in \mathcal{V}_n$ denote the kth and ℓth nodes (objects) in image $\mathbf{I}_n$. The adjacency matrix $\mathbf{A}_n$ is determined by the dataset, based on the provided object annotations and edge-relation annotations.

In the graph convolution model $\mathcal{F}_{\theta_g}$, we operate on the graph structure $\mathcal{T}_n = (\mathcal{V}_n, \mathcal{E}_n)$ to perform convolution over its nodes $\mathcal{V}_n$ and edges $\mathcal{E}_n$. The GCN aggregates information from neighboring nodes, allowing each object's representation to be updated by incorporating the spatial and relational information of nearby objects.

Specifically, the GCN updates each node's feature as

$$\mathbf{h}_{n,k}^{(l+1)} = \sigma\left(\sum_{\ell \in \mathcal{N}(k)} \frac{[\mathbf{A}_n]_{k\ell}}{\sqrt{d_k d_\ell}} \mathbf{W}^{(l)} \mathbf{h}_{n,\ell}^{(l)}\right) \tag{5}$$

where $\mathbf{h}_{n,k}^{(l)}$ denotes the feature vector of the kth node at layer l, $\mathcal{N}(k)$ is the set of neighboring nodes of the kth node, $\mathbf{W}^{(l)}$ is a learnable weight matrix at layer l, d_k is the degree of node k, and $\sigma(\cdot)$ is a nonlinear activation function such as ReLU. The normalization term $([\mathbf{A}_n]_{k\ell})/((d_k d_\ell)^{1/2})$ corresponds to the symmetric normalization of the adjacency matrix, ensuring numerical stability and balanced aggregation.

After L layers, we obtain the set of node embeddings

$$\mathbf{H}_n = \mathcal{F}_{\theta_g}(\mathcal{T}_n) \tag{6}$$

where L is the total number of GCN layers, $\mathbf{H}_n = \{\mathbf{h}_{n,k}^{(L)}\}_{k=1}^{K}$ denotes the set of node embeddings at the final layer, $\mathbf{h}_{n,k}^{(L)} \in \mathbb{R}^d$ is the embedding of the kth node, K is the number of nodes (objects) in image $\mathbf{I}_n$, and $\mathbf{H}_n \in \mathbb{R}^{K \times d}$ denotes the matrix stacking all node embeddings.

C. Caption Generation

In order to generate the final caption from graph-encoded object features, we develop a graph decoding model $\mathcal{M}_{\theta_m}$, as

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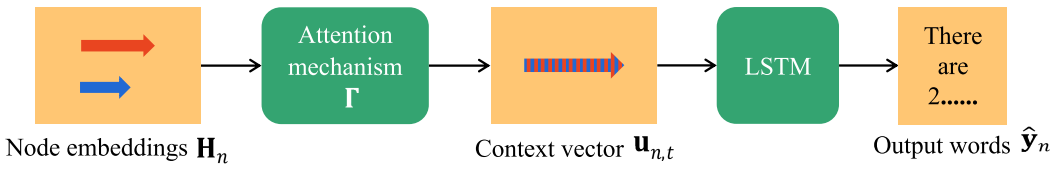


Fig. 3. Graph decoding. The red-blue alternating arrow in the context vector model denotes the vector produced from the node embeddings after the attention mechanism model.

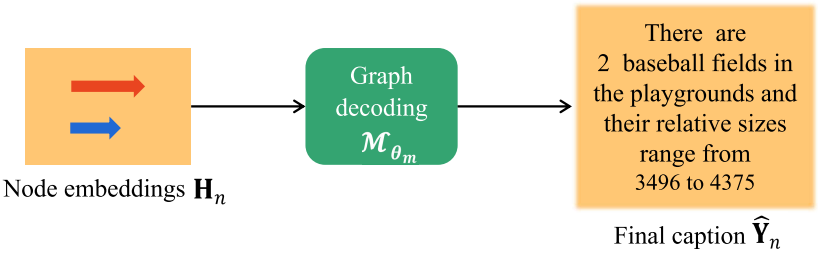


Fig. 4. Caption generation.

illustrated in Fig. 3 and develop a caption generation module, as illustrated in Fig. 4. It contains the graph decoding model $\mathcal{M}_{\theta_m}$. The node embeddings $\mathbf{H}_n$ derived from the GCN are input into the graph decoding model $\mathcal{M}_{\theta_m}$, and the final caption $\hat{\mathbf{Y}}_n$ is generated.

Specifically, in the graph decoding model $\mathcal{M}_{\theta_m}$ we use a long short-term memory (LSTM) network with learnable parameters $\mathbf{W}_h$ and $\mathbf{W}_s$ to generate output words $\hat{\mathbf{y}}_n$ and the final caption $\hat{\mathbf{Y}}_n$ from the node embeddings $\mathbf{H}_n$.

The node embeddings $\mathbf{H}_n = \{\mathbf{h}_{n,k}^{(L)}\}_{k=1}^{K}$ obtained from the GCN are processed by an attention mechanism Γ. Let $\mathbf{h}_{n,k}$ denote the embedding of the kth object, and let $\mathbf{q}_t$ be the hidden state of the LSTM decoder at time step t. We define $\mathbf{v}$ as a learnable vector used to compute attention scores. The attention weights $\alpha_{t,k}$ and the context vector $\mathbf{u}_{n,t}$ (which is represented by the red-blue alternating arrow in Fig. 4) are computed as

$$\alpha_{t,k} = \frac{\exp\left(\eta_{t,k}\right)}{\sum_{j=1}^{K}\exp\left(\eta_{t,j}\right)}$$
$$\eta_{t,k} = \mathbf{v}^{\top}\tanh\left(\mathbf{W}_h\mathbf{h}_{n,k} + \mathbf{W}_s\mathbf{q}_{t-1}\right) \quad (7)$$
$$\mathbf{u}_{n,t} = \sum_{k=1}^{K}\alpha_{t,k}\mathbf{h}_{n,k} \quad (8)$$

where $\eta_{t,k}$ is the attention score of the kth node at decoding step t, $\alpha_{t,k}$ is the normalized attention weight obtained via softmax, and $\mathbf{u}_{n,t}$ denotes the context vector at decoding step t for image $\mathbf{I}_n$.

The context vector $\mathbf{u}_{n,t}$ is fed into the LSTM decoder together with the embedding of the previously generated word. Specifically, let $\mathbf{w}_{t-1} \in \mathbb{R}^d$ denote the word embedding of the word y_{t-1}, obtained from an embedding matrix $\mathbf{W}_e \in \mathbb{R}^{d\times V}$. At each step t, the LSTM updates its hidden state as

$$\mathbf{q}_t = \text{LSTM}\left(\left[\mathbf{w}_{t-1}; \mathbf{u}_{n,t}\right], \mathbf{q}_{t-1}\right) \quad (9)$$

where $[\cdot;\cdot]$ denotes vector concatenation.

The probability distribution over the vocabulary is then computed as

$$P\left(y_t \mid y_{<t}, \mathbf{H}_n\right) = \text{Softmax}\left(\mathbf{W}_o\mathbf{q}_t + \mathbf{b}_o\right) \quad (10)$$

where $y_{<t} = (y_1, \ldots, y_{t-1})$ denotes all previously generated words before time step t, $\mathbf{W}_o \in \mathbb{R}^{V\times d_s}$ projects the hidden state to the vocabulary space, $\mathbf{b}_o \in \mathbb{R}^V$ is the output bias vector, and $\mathbf{H}_n = \{\mathbf{h}_{n,k}^{(L)}\}_{k=1}^{K}$ denotes the GCN-derived node embeddings.

The final caption sequence is generated autoregressively until the special end-of-sequence token ⟨EOS⟩ is produced

$$\hat{\mathbf{Y}}_n = \mathcal{M}_{\theta_m}\left(\mathbf{H}_n\right) \quad (11)$$

where $\hat{\mathbf{Y}}_n = (y_1, y_2, \ldots, y_T)$ denotes the predicted caption, and $\mathcal{M}_{\theta_m}$ is the graph decoding model parameterized by θ_m, which decodes the graph-based node embeddings $\mathbf{H}_n$.

D. Overall Framework and Loss Function

Given an input image $\mathbf{I}_n$, we first apply a Mask R-CNN object confirmation module $\mathcal{D}_{\theta_d}$ to segment objects and obtain their class labels $c_{n,k}$. We calculate the corresponding relative size $s_{n,k}$, and normalize it to obtain the object relative size $\hat{s}_{n,k}$. The resulting feature pair $(c_{n,k}, \hat{s}_{n,k})$ serves as the initial representation for each node. The confirmation performance is supervised using a composite loss function that includes the standard Mask R-CNN loss $\mathcal{L}_{\text{det}}$ and a mean squared error loss $\mathcal{L}_{\text{rs}}$ for relative size (abbreviated as "rs") estimation

$$\mathcal{L}_1 = \mathcal{L}_{\text{det}} + \lambda_{\text{rs}}\mathcal{L}_{\text{rs}}. \quad (12)$$

Subsequently, we develop a graph construction model $\mathcal{G}_n = (\mathcal{V}_n, \mathcal{E}_n)$, where each node $v_k \in \mathcal{V}_n$ corresponds to an object and embeds the feature vector $\mathbf{h}_{n,k}^{(0)} = [e(c_{n,k}), \hat{s}_k]$. Edges $\mathcal{E}_n$ are formed by evaluating spatial relationships between objects. A multilayer graph convolutional network $\mathcal{F}_{\theta_g}$ is then applied to iteratively update node representations based on neighborhood aggregation. The quality of graph structure and node embeddings is encouraged through auxiliary losses, including a binary cross-entropy loss $\mathcal{L}_{\text{adj}}$ for adjacency supervision

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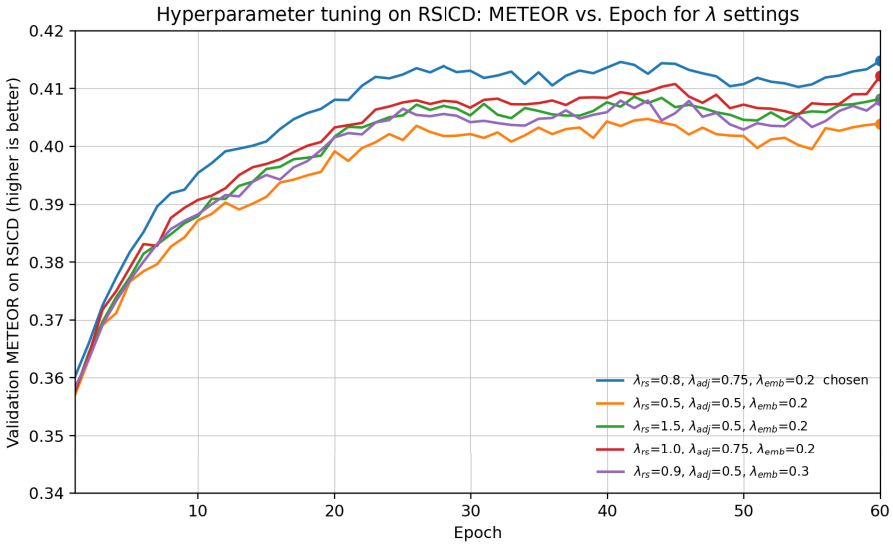


Fig. 5. Hyperparameter tuning of the weighting coefficients on the RSICD dataset. The curves show the validation Meteor performance over 60 epochs under different settings, where $\lambda_{\text{rs}} = 0.8$, $\lambda_{\text{adj}} = 0.75$, and $\lambda_{\text{emb}} = 0.2$ achieve the highest results.

(abbreviated as "adj") and an optional contrastive embedding loss (abbreviated as "emb") $\mathcal{L}_{\text{emb}}$

$$\mathcal{L}_2 = \lambda_{\text{adj}}\mathcal{L}_{\text{adj}} + \lambda_{\text{emb}}\mathcal{L}_{\text{emb}}. \tag{13}$$

Finally, the enhanced node features are passed into an attention-based LSTM decoder $\mathcal{Y}_{\theta_y}$ to sequentially generate the image caption $\hat{\mathbf{Y}}_n$. At each decoding step t, attention weights are computed over all graph nodes to produce a context vector $\mathbf{u}_{n,t}$, which, combined with the previous word embedding, updates the LSTM state for predicting the next word. The decoder is optimized using a negative log-likelihood loss (abbreviated as "cap") $\mathcal{L}_{\text{cap}}$

$$\mathcal{L}_3 = \mathcal{L}_{\text{cap}}. \tag{14}$$

The weighting coefficients λ_{rs}, λ_{adj}, and λ_{emb} in the total loss function were determined through tuning on the validation subset of the remote sensing image captioning dataset (RSICD) dataset, which we will present in Section IV-B. As illustrated in Fig. 5, the configuration of $\lambda_{\text{rs}} = 0.8$, $\lambda_{\text{adj}} = 0.75$, and $\lambda_{\text{emb}} = 0.2$ achieved the highest validation Meteor value, which we will present in Section III-A. To jointly optimize the entire framework, we define the total loss as a weighted sum of the three module-specific losses

$$\mathcal{L}_{\text{total}} = \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3. \tag{15}$$

III. New Evaluation Metric of SizeNum-Meteor

In this section, we design a new evaluation metric SizeNum-Meteor. We first review the traditional Meteor metric, which is widely used in image captioning for measuring semantic and linguistic quality. We then introduce our proposed SizeNum-Meteor metric, which extends Meteor by explicitly incorporating object count and relative size information to better suit the requirements of remote sensing captioning.

A. Traditional Meteor

The Meteor (Metric for Evaluation of Translation with Explicit ORdering) metric [30] was originally proposed for evaluating machine translation, and has been widely adopted in image captioning tasks. Meteor improves the evaluation by incorporating stemming, synonym matching, and word order alignment. It computes a harmonic mean of unigram precision and recall, with an additional penalty term for word order deviations. Formally, unigram precision P and recall R are defined as

$$P = \frac{m}{w_t}, \qquad R = \frac{m}{w_r} \tag{16}$$

where m is the number of matched unigrams, w_t is the number of unigrams in the candidate caption, and w_r is the number of unigrams in the reference caption. The harmonic mean is given by

$$F_\alpha = \frac{PR}{\alpha P + (1-\alpha)R} \tag{17}$$

and the penalty term is computed as

$$\text{Penalty} = \gamma\left(\frac{ch}{m}\right)^{\beta} \tag{18}$$

where ch is the number of contiguous matched chunks, and γ and β are hyperparameters. The final Meteor score is obtained as

$$\text{Meteor} = \big(1 - \text{Penalty}\big)\, F_\alpha. \tag{19}$$

B. Novel SizeNum-Meteor

While Meteor effectively captures semantic similarity and linguistic quality, it fails to explicitly evaluate the correctness of quantitative information such as object counts and relative sizes, which are critical in remote sensing captioning. To address this limitation, we propose SizeNum-Meteor, a custom metric designed to evaluate the accuracy of object count and relative size information in generated captions. It extends the original Meteor score by incorporating penalties for numerical

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Fig. 6. Examples from remote sensing image datasets with segmentation labels.

discrepancies in object count and relative size between the generated caption and the reference. The metric is defined as

$$\text{SizeNum-Meteor} = \frac{\text{Meteor}}{\exp\left(|\text{count}_{\text{GT}} - \text{count}_{\text{Ref}}| + |\text{size}_{\text{GT}} - \text{size}_{\text{Ref}}|\right)} \tag{20}$$

where count_{GT} and size_{GT} denote the ground-truth object count and relative size, while $\text{count}_{\text{Ref}}$ and size_{Ref} are the values extracted from the generated caption.

IV. Relative Size-Annotated Remote Sensing Captioning Datasets

In this section, we introduce baseline datasets that can be used for validating the proposed strategy. To provide pixel-level object masks for pretraining and supervising the object confirmation module, we employ remote sensing image datasets with segmentation labels. To serve as widely adopted benchmarks for conventional captioning tasks, we adopt remote sensing image captioning datasets with preliminary caption labels. To enable quantitatively enhanced captioning, we construct a relative size-annotated remote sensing captioning benchmark.

A. Remote Sensing Image Datasets With Segmentation Labels

Dataset for Instance Segmentation in Aerial Images (iSAID)[1]: The iSAID dataset is a large-scale instance segmentation benchmark for aerial images, derived from the DOTA dataset [31], [32]. It contains 2806 high-resolution images with a total of 6 55 451 object instances, annotated across 15 categories including plane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, large vehicle, small vehicle, helicopter, roundabout, soccer ball field, and swimming pool. The images have resolutions ranging from 800 × 800 to 4000 × 4000 pixels, covering diverse urban and industrial scenes with dense and small-object distributions. iSAID is widely used for remote sensing object instance segmentation. Examples are shown in Fig. 6.

[1]The iSAID dataset is openly available at https://captain-whu.github.io/iSAID/

B. Remote Sensing Image Captioning Datasets With Preliminary Caption Labels

UC Merced Land Use Dataset[2]*:* The UC Merced (UCM) dataset is created by the University of California, Merced. It contains 2100 images, categorized into 21 different classes, with each class containing 100 images. The images have a resolution of 256 × 256 pixels, covering different scene categories such as plane, baseball field, beach, and more. UCM dataset is widely used for remote sensing image captioning.

Sydney Urban Objects (Sydney)[2]*:* The Sydney dataset is created by the University of Sydney. It contains 613 images, categorized into 7 different classes, with each class containing about 100 images. The images have a resolution of 500 × 500 pixels, covering different scene categories such as airports, oceans, and factories. Sydney dataset is widely used for remote sensing image captioning.

Remote Sensing Image Captioning Dataset[2]*:* The RSICD dataset is created by Northwestern Polytechnical University. It contains 10 921 images, categorized into 30 different classes, with each class containing about 360 images. The images have a resolution of 224 × 224 pixels, covering different scene categories such as farmland, urban areas, forests, and water bodies. RSICD is widely used for remote sensing image captioning.

Examples from the UCM, Sydney, and RSICD datasets are shown in Fig. 7, and the detailed statistics of these three datasets are summarized in Table II.

C. Remote Sensing Image Captioning Datasets With Caption Labels Incorporating Object Count and Relative Size Information

UCM-N-S[3]*:* The UCM-N-S dataset is constructed based on the UCM dataset, in which each image is annotated with captions that explicitly describe both the number and the relative size of objects. Captions reflect fine-grained details such as "2 storage tanks" or "with relative sizes ranging from 9362 to 11 522."

Sydney-N-S[3]*:* The Sydney-N-S dataset is constructed based on the Sydney dataset, in which each image is annotated

[2]The UCM, Sydney, and RSICD datasets are openly available at https://github.com/201528014227051/RSICD_optimal

[3]The UCM-N-S, Sydney-N-S, and RSICD-N-S datasets are openly available at https://github.com/zihaoni/SizeCaption

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Caption: There are some plans in the airport.



Caption: There are some storage tanks in the factory.



Caption: There is a plane in the airport.

Fig. 7. Examples from remote sensing image captioning datasets with preliminary caption labels.

TABLE II
SUMMARY OF UCM, SYDNEY, AND RSICD DATASETS

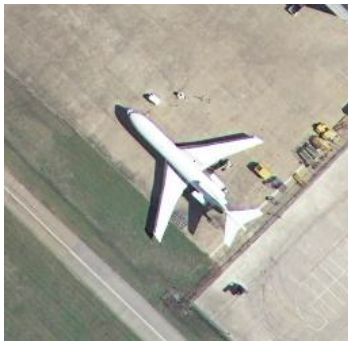
Dataset	Image count	Number of categories	Image resolution
UCM	2,100	21	256 × 256 pixels
Sydney	613	7	500 × 500 pixels
RSICD	10,921	30	224 × 224 pixels



Caption: There are 2 bridges on the river and their relative sizes range from 9362 to 11522.



Caption: There is 1 playground in the field and it's relative size is 12596.



Caption: There is 1 plane in the airport and it's relative size is 7864.

Fig. 8. Examples from remote sensing image captioning datasets with caption labels incorporating object count and relative size information.

with captions that explicitly describe both the number and the relative size of objects. Examples include "3 baseball fields" or "its relative size is 12 596."

RSICD-N-S[3]*:* The RSICD-N-S dataset is constructed based on the original RSICD dataset, in which each image is annotated with captions that explicitly describe both the number and the relative size of objects. Captions include phrases such as "1 house" or "ranging from 2936 to 5861."

Examples from UCM-N-S, Sydney-N-S and RSICD-N-S are shown in Fig. 8.

V. EXPERIMENTS

In this section, we conduct experiments to validate the effectiveness of our proposed framework. We introduce the experimental platform and we evaluate our method against a set of state-of-the-art (SOTA) captioning models, using both standard metrics (Bleu, Rouge, Cider, semantic propositional image caption evaluation (SPICE), Meteor) and our proposed SizeNum-Meteor. Finally, we present ablation analysis to investigate the individual contributions of each module in our framework.

A. Experimental Platform

We conduct all experiments on a workstation equipped with an Intel Xeon Gold 6230 CPU at 2.10 GHz, 256 GB RAM, and two NVIDIA RTX 3090 GPUs. The software environment includes Ubuntu 20.04, Python 3.9, PyTorch 1.13, and CUDA 11.6.

B. Training

The object confirmation model in the object confirmation and relative size estimation module is pretrained on the remote sensing image dataset iSAID with segmentation labels. The

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TABLE III

EVALUATION RESULTS ON THE UCM AND UCM-N-S DATASETS. THE BOLD NUMBERS IN THE TABLES MARK THE HIGHEST SCORES FOR EACH COLUMN. "-" INDICATES THAT THE CORRESPONDING METRIC IS NOT APPLICABLE

Framework	Bleu1	Bleu2	Bleu3	Bleu4	Rouge	Cider	Spice	Meteor	SizeNum-Meteor
TrTr-CMR	0.8351	0.7265	0.6410	0.5652	0.7641	3.0210	0.4023	0.4156	–
MST	0.8329	0.7942	0.6280	0.6207	0.7598	3.2764	0.3981	0.4120	–
RSIC-GMamba	0.9280	0.8982	0.8705	0.8463	**0.8965**	**4.6012**	0.5852	0.5603	–
M-T-C	0.8625	0.8012	0.7564	0.7123	0.8239	3.4821	0.4721	0.5014	–
MVP	0.8902	0.8430	0.7996	0.7581	0.8449	3.7824	0.5283	0.4901	–
EfficientNet	0.7949	0.7140	0.6542	0.6018	0.7452	3.0648	0.4212	0.4160	–
TR-FRIC	0.8666	0.8139	0.7736	0.7222	0.8156	3.6749	0.5168	0.4826	–
FRR	0.8530	0.7970	0.7485	0.7050	0.8115	3.4912	0.4705	0.4668	–
Ours	**0.9285**	**0.9198**	**0.8891**	**0.8627**	0.8912	4.3954	**0.6189**	**0.6291**	**0.0547**

graph construction and graph convolution modules are trained jointly with the caption decoder on our UCM-N-S, Sydney-N-S, and RSICD-N-S datasets. We present the experimental platform and compare the experimental results of our proposed framework and SOTA methods including TrTr-CMR [15], MST [16], RSIC-GMamba [17], M-T-C [18], MVP [19], EfficientNet [20], TR-FRIC [21], and FRR [22].

C. Evaluation Metrics

To comprehensively assess the performance of our remote sensing image captioning framework, we employ a suite of widely adopted evaluation metrics that capture different aspects of caption quality. These metrics include lexical overlap [bilingual evaluation understudy (Bleu), recall-oriented understudy for gisting evaluation (Rouge)], semantic alignment (Cider, SPICE), and linguistic fluency (Meteor). However, standard metrics often overlook the correctness of numerical and quantitative information such as object count and size, which are critical in many remote sensing applications. To address this gap, we propose SizeNum-Meteor, a custom metric specifically designed to evaluate the numerical consistency between generated captions and ground-truth annotations.

1) *Bilingual Evaluation Understudy [33]:* Bleu measures the degree of n-gram overlap between the generated caption and one or more reference captions. It primarily captures surface-level fluency.
2) *Recall-Oriented Understudy for Gisting Evaluation [34]:* Rouge evaluates the longest common subsequence (LCS) between the generated and reference captions, emphasizing sentence-level structural similarity.
3) *Consensus-based Image Description Evaluation [35]:* Cider computes n-gram similarity between generated and reference captions. It emphasizes the alignment with consensus words from multiple human references.
4) *Semantic Propositional Image Caption Evaluation [36]:* SPICE parses both reference and generated captions into scene graphs containing objects, attributes, and relationships, and computes an $F1$ score over matching tuples. It is designed to measure semantic correctness rather than surface matching.
5) *Meteor*: Presented in Section III-A.
6) *SizeNum-Meteor (Ours)*: Presented in Section III-B.

TABLE IV

EVALUATION RESULTS ON THE SYDNEY AND SYDNEY-N-S DATASETS

Framework	Bleu1	Bleu2	Bleu3	Bleu4	Rouge	Cider	Spice	Meteor	SizeNum-Meteor
TrTr-CMR	0.8440	0.7169	0.6198	0.5391	0.7422	2.3520	0.3987	0.3921	–
MST	0.8067	0.7182	0.6031	0.5910	0.7388	2.3651	0.3940	0.3895	–
RSIC-GMamba	0.8599	0.8314	0.7449	**0.7298**	**0.8177**	**3.4915**	0.4995	0.4751	–
M-T-C	0.7846	0.7050	0.6433	0.5954	0.7456	2.9162	0.4634	0.4192	–
MVP	0.8159	0.7428	0.6796	0.6189	0.7443	2.6099	0.4831	0.4113	–
EfficientNet	0.7175	0.6282	0.5482	0.4817	0.6725	2.1746	0.4110	0.3617	–
TR-FRIC	0.8299	0.7454	0.6623	0.5815	0.7292	2.5652	0.4529	0.4318	–
FRR	0.7822	0.6992	0.6252	0.5561	0.7176	2.4824	0.4666	0.4007	–
Ours	**0.8602**	**0.8461**	**0.7498**	0.6994	0.7524	2.5523	**0.6111**	**0.4785**	**0.0449**

TABLE V

EVALUATION RESULTS ON THE RSICD AND RSICD-N-S DATASETS

Framework	Bleu1	Bleu2	Bleu3	Bleu4	Rouge	Cider	Spice	Meteor	SizeNum-Meteor
TrTr-CMR	0.6362	0.4053	0.2789	0.2050	0.5020	2.7721	0.3901	0.2513	–
MST	0.8273	0.6938	0.5532	0.5080	**0.7412**	2.7415	0.3964	0.3422	–
RSIC-GMamba	0.7068	0.6024	0.5261	0.4651	0.6314	2.8103	0.4819	0.3486	–
M-T-C	0.6661	0.4973	0.3889	0.3248	0.5005	2.8527	0.4698	0.2897	–
MVP	0.7971	0.6952	0.6108	0.5372	0.6927	**2.9231**	0.4947	0.3739	–
EfficientNet	0.6363	0.4615	0.3552	0.2886	0.4782	2.7899	0.4031	0.2621	–
TR-FRIC	0.7978	0.6842	0.5973	0.5215	0.6941	2.8761	0.5116	0.3081	–
FRR	0.7831	0.6681	0.5770	0.5048	0.6739	2.8447	0.4681	0.3677	–
Ours	**0.8164**	**0.7246**	**0.6942**	**0.6221**	0.7345	2.6118	**0.5287**	**0.4131**	**0.0403**

TABLE VI

EVALUATION RESULTS ON THE UCM-N-S DATASET

Framework	Bleu1	Bleu2	Bleu3	Bleu4	Rouge	Cider	Spice	Meteor	SizeNum-Meteor
TrTr-CMR	0.6156	0.5091	0.4220	0.3469	0.5442	2.1050	0.2987	0.1978	–
MST	0.6270	0.5790	0.4230	0.4160	0.5523	2.5490	0.3010	0.2250	–
RSIC-GMamba	0.7140	0.6842	0.6572	0.6318	0.6822	3.6165	0.3817	0.3565	–
M-T-C	0.6460	0.5840	0.5390	0.4930	0.6070	2.7080	0.3124	0.3070	–
MVP	0.7005	0.6521	0.6090	0.5689	0.6541	3.0405	0.3478	0.3097	–
EfficientNet	0.6043	0.5238	0.4638	0.4109	0.5544	2.3826	0.2766	0.2362	–
TR-FRIC	0.6760	0.6237	0.5833	0.5316	0.6152	2.9430	0.3363	0.3023	–
FRR	0.6634	0.6068	0.5587	0.5158	0.6118	2.7400	0.3298	0.2865	–
Ours	**0.9285**	**0.9198**	**0.8891**	**0.8627**	**0.8912**	**4.3954**	**0.6189**	**0.6291**	**0.0547**

D. Evaluation Results

To evaluate the effectiveness of our proposed framework, we conducted comprehensive experiments on two groups of datasets: 1) remote sensing image captioning datasets with preliminary caption labels (UCM, Sydney, and RSICD); and 2) remote sensing image captioning datasets with caption labels incorporating object count and relative size information (UCM-N-S, Sydney-N-S, and RSICD-N-S). The evaluation was performed in two experimental settings:

Setting A (Tables III–V): Existing SOTA methods were tested on UCM, Sydney, and RSICD, while our proposed framework was evaluated on UCM-N-S, Sydney-N-S, and RSICD-N-S.

Setting B (Tables VI–VIII): Both existing SOTA methods and our proposed framework were trained and tested on UCM-N-S, Sydney-N-S, and RSICD-N-S.

As shown in Tables III–V, our proposed framework achieves better performance across almost all metrics. For example, on UCM, our proposed framework achieves Bleu4 = 0.8627, Rouge = 0.8912, and SizeNum-Meteor = 0.0547. On Sydney, our proposed framework achieves Bleu4 = 0.6994, SPICE = 0.6111, and SizeNum-Meteor = 0.0449, again higher than competing methods. On RSICD, our proposed framework

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Final caption: There are 3 baseball fields in the playgrounds and their relative sizes range from 2832 to 3357.



Final caption: There are 2 storage tanks in the field and their relative sizes range from 3310 to 3311.



Final caption: There is 1 house in the field and it's relative size is 2355.

Fig. 9. Samples.

TABLE VII

EVALUATION RESULTS ON THE SYDNEY-N-S DATASET

Framework	Bleu1	Bleu2	Bleu3	Bleu4	Rouge	Cider	Spice	Meteor	SizeNum-Meteor
TrTr-CMR	0.5970	0.4694	0.3802	0.2999	0.5020	1.7428	0.2211	0.1803	–
MST	0.5510	0.4730	0.3620	0.3290	0.4983	1.7060	0.2105	0.1730	–
RSIC-GMamba	0.6468	0.5965	0.5505	0.4848	0.5824	1.8157	0.2944	0.2681	–
M-T-C	0.5580	0.4690	0.4060	0.3490	0.5110	1.7200	0.2280	0.2010	–
MVP	0.5954	0.5232	0.4593	0.3782	0.5345	1.8949	0.2833	0.2207	–
EfficientNet	0.4871	0.3979	0.3188	0.2512	0.4520	1.5430	0.1955	0.1706	–
TR-FRIC	0.5995	0.5041	0.4212	0.3404	0.5097	1.8323	0.2622	0.2211	–
FRR	0.5615	0.4794	0.3957	0.3169	0.4967	1.7508	0.2043	0.2000	–
Ours	**0.8602**	**0.8461**	**0.7498**	**0.6994**	**0.7524**	**2.5523**	**0.6111**	**0.4785**	**0.0449**

TABLE VIII

EVALUATION RESULTS ON THE RSICD-N-S DATASET

Framework	Bleu1	Bleu2	Bleu3	Bleu4	Rouge	Cider	Spice	Meteor	SizeNum-Meteor
TrTr-CMR	0.2901	0.1437	0.0871	0.0432	0.1895	1.6218	0.1012	0.0999	–
MST	0.3638	0.2804	0.2042	0.1338	0.3073	1.9513	0.1943	0.0963	–
RSIC-GMamba	0.4730	0.3610	0.2410	0.1760	0.3066	1.6950	0.1543	0.1100	–
M-T-C	0.3210	0.1520	0.0950	0.0480	0.1800	1.7070	0.1120	0.0830	–
MVP	0.4480	0.3656	0.2700	0.1969	0.3621	1.9048	0.2131	0.1234	–
EfficientNet	0.3057	0.1505	0.0961	0.0471	0.1674	1.6888	0.0970	0.0712	–
TR-FRIC	0.4471	0.3449	0.2581	0.1726	0.3655	1.9886	0.2224	0.1169	–
FRR	0.4336	0.3279	0.2374	0.1642	0.3430	1.9436	0.2033	0.1172	–
Ours	**0.8164**	**0.7426**	**0.6942**	**0.6221**	**0.7345**	**2.6118**	**0.5287**	**0.4131**	**0.0403**

achieves Bleu4 = 0.6221, Meteor = 0.4131, and SizeNum-Meteor = 0.0403, clearly outperforming prior work.

As shown in Tables VI–VIII, our proposed framework achieves better results when all models were trained and tested on UCM-N-S, Sydney-N-S, and RSICD-N-S. On UCM-N-S, our proposed framework achieves Bleu4 = 0.8627 and SizeNum-Meteor = 0.0547; on Sydney-N-S, our proposed framework achieves Bleu4 = 0.6994 and SizeNum-Meteor = 0.0449; and on RSICD-N-S, our proposed framework achieves Bleu4 = 0.6221 and SizeNum-Meteor = 0.0403.

These results confirm the robustness and superiority of our proposed framework in accurately capturing object count and relative size information for remote sensing image captioning.

To analyze the improvement more concretely, the incorporation of relative size in our proposed framework notably enhances metrics related to semantic richness and numerical precision, such as Meteor and SizeNum-Meteor. Compared with existing methods that rely solely on visual and positional features, our proposed framework integrates relative object size into the graph construction and decoding stages, thereby enabling a more structured understanding of remote sensing images. For instance, the increase in the Meteor score across datasets (Tables III–VIII) indicates that the final captions are not only linguistically fluent but also better able to express object relative size information.

Furthermore, the advantage of our proposed framework becomes more pronounced in the UCM-N-S, Sydney-N-S, and RSICD-N-S datasets, where object count and object relative size annotations are explicitly available. This suggests that our proposed framework effectively exploits the enriched annotations to learn object relationship and size information. Our proposed framework is able to generate captions with more accurate object counts and object relative sizes, resulting in better semantic consistency, which is rarely achieved by existing methods.

To further illustrate the quality of the final captions, Fig. 9 provides several qualitative examples. The final captions generated by our model successfully describe not only the object classes (e.g., "baseball fields," "planes," "storage tanks") but also accurately reflect their count and relative size range.

E. Ablation Analysis

We design a series of ablation experiments to evaluate the contribution of each key component in our proposed framework. Table IX summarizes the settings and objectives of each variant. The Baseline model removes all structure- and quantity-aware modules, while subsequent variants reintroduce specific components one at a time. By comparing their performances with the Full-Model, we can isolate the effect of object segmentation, graph construction, graph convolution, and an attention mechanism on caption generation quality,

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TABLE IX
ABLATION EXPERIMENT CONFIGURATIONS AND THEIR PURPOSES

Model	Configuration	Purpose
Baseline	The Baseline configuration does not use the object confirmation module, the graph construction model, or relative size information. It extracts features using a standard CNN and generates captions with an LSTM.	The Baseline configuration serves as a basic reference to assess captioning without structural or quantitative guidance.
No-Segmentation	The No-Segmentation configuration uses ground-truth object masks and class labels instead of the outputs from the object confirmation module, thus skipping the object confirmation process.	The No-Segmentation configuration evaluates the impact of the object confirmation module by eliminating its potential errors and measuring the upper-bound performance.
No-Graph	The No-Graph configuration removes the graph construction model. Object features (class + relative size) are concatenated directly without modeling spatial or semantic relationships between objects.	The No-Graph configuration assesses the importance of graph construction in capturing inter-object relations and enhancing representation.
No-GCN	The No-GCN configuration constructs the graph but omits the graph convolutional network (GCN). Node features are used directly without neighborhood aggregation.	The No-GCN configuration tests the effectiveness of the GCN in enriching node representations through relational context.
No-Attention	The No-Attention configuration removes the attention mechanism in the decoder. All node features are averaged before being fed into the LSTM decoder.	The No-Attention configuration verifies the role of attention in focusing on salient targets and generating accurate relative size- and count-aware captions.
Full-Model	The Full-Model includes all components: Mask R-CNN-based object confirmation, relative size estimation, graph construction, graph convolution (GCN), and the attention-based LSTM decoder.	The Full-Model demonstrates the combined effect of all modules in generating high-quality, size-sensitive captions.

TABLE X
ABLATION RESULTS ON THE UCM-N-S DATASET

Model	Bleu1	Bleu2	Bleu3	Bleu4	Rouge	Cider	Spice	Meteor	SizeNum-Meteor
Baseline	0.5903	0.4705	0.3682	0.2891	0.5521	1.8012	0.2761	0.2422	–
No-Segmentation	0.8924	0.8803	0.8501	0.8185	0.8612	3.9451	0.5740	0.5902	0.0485
No-Graph	0.8120	0.7815	0.7204	0.6411	0.7655	3.0123	0.4631	0.4528	–
No-GCN	0.8457	0.8146	0.7529	0.6782	0.7923	3.2245	0.5012	0.4783	0.0365
No-Attention	0.8675	0.8422	0.7910	0.7136	0.8119	3.4866	0.5290	0.5019	0.0407
Full-Model (Ours)	**0.9285**	**0.9198**	**0.8891**	**0.8627**	**0.8912**	**4.3954**	**0.6189**	**0.6291**	**0.0547**

TABLE XI
ABLATION RESULTS ON THE SYDNEY-N-S DATASET

Model	Bleu1	Bleu2	Bleu3	Bleu4	Rouge	Cider	Spice	Meteor	SizeNum-Meteor
Baseline	0.5311	0.4286	0.3371	0.2594	0.4723	1.5018	0.2440	0.2087	–
No-Segmentation	0.8252	0.8120	0.7215	0.6689	0.7312	2.4015	0.6853	0.5562	0.0402
No-Graph	0.7529	0.7268	0.6347	0.5128	0.6685	1.9823	0.5761	0.4415	–
No-GCN	0.7804	0.7559	0.6624	0.5492	0.6911	2.1132	0.6054	0.4703	0.0311
No-Attention	0.8046	0.7802	0.6895	0.5881	0.7128	2.2764	0.6327	0.4902	0.0360
Full-Model (Ours)	**0.8602**	**0.8461**	**0.7498**	**0.6994**	**0.7524**	**2.5523**	**0.6111**	**0.4785**	**0.0449**

especially under the constraint of incorporating count and size information.

Tables X–XII present the ablation analysis on UCM-N-S, Sydney-N-S, and RSICD-N-S. The results show that the Baseline configuration yields the lowest scores across all datasets. Removing graph construction or graph convolution causes substantial performance degradation, and the negative impact becomes more pronounced as dataset complexity increases. The absence of attention also reduces performance, though the drop is relatively smaller compared with graph-related components.

Regarding SizeNum-Meteor, configurations lacking size- or structure-related modules fail to capture reliable numerical information, leading to missing or heavily reduced scores in this metric. The graph-related modules, particularly the graph construction and GCN components, make the most significant contribution to the overall performance improvement. By jointly modeling spatial relationships among objects and their relative sizes, these modules enable the framework to capture spatial relationships and interobject relations more effectively and to incorporate object relative size information into caption generation. In contrast, the

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TABLE XII
ABLATION RESULTS ON THE RSICD-N-S DATASET

Model	Bleu1	Bleu2	Bleu3	Bleu4	Rouge	Cider	Spice	Meteor	SizeNum-Meteor
Baseline	0.4820	0.3624	0.2701	0.1902	0.4302	1.2324	0.2218	0.1865	–
No-Segmentation	0.7655	0.6928	0.6401	0.5742	0.6985	2.2114	0.4821	0.4518	0.0347
No-Graph	0.6704	0.5912	0.5286	0.4012	0.6097	1.7432	0.4016	0.3722	–
No-GCN	0.7059	0.6238	0.5629	0.4430	0.6385	1.8964	0.4298	0.3981	0.0275
No-Attention	0.7342	0.6520	0.5925	0.4903	0.6612	2.0159	0.4517	0.4210	0.0310
Full-Model (Ours)	**0.8164**	**0.7426**	**0.6942**	**0.6221**	**0.7345**	**2.6118**	**0.5287**	**0.4131**	**0.0403**

Full-Model consistently preserves strong SizeNum-Meteor values, confirming that the joint use of object confirmation, graph reasoning, and attention is essential for generating captions that are both semantically accurate and quantitatively faithful.

Compared with the ablated variants, the proposed Full-Model achieves consistent gains in Bleu4, Cider, and Meteor, with the advantage being particularly evident on RSICD-N-S. These results indicate that explicitly modeling relationships between objects via graph construction and graph convolution is crucial for generating semantically coherent captions.

When the GCN or the attention mechanism is removed, we observe moderate but clear performance drops, confirming that relational aggregation and focus selection are both essential for precise caption generation. In particular, the decline in SizeNum-Meteor under the No-Attention setting shows that adaptive focus helps preserve quantitative correctness when describing object relative size.

Overall, the ablation analysis verifies that each component contributes meaningfully, and their integration forms a balanced framework that generates captions which are contextually coherent and quantitatively reliable with respect to object count and object relative size.

VI. CONCLUSION

In this work, we proposed a size-aware graph embedding framework for remote sensing image captioning. First, existing methods often neglect explicit modeling of object size-an essential quantitative attribute in geo-observation, so we designed a framework with object confirmation and size estimation to generate captions that are both precise and informative. Second, conventional metrics emphasize fluency and semantic similarity but ignore quantitative correctness, to address this, we introduced SizeNum-Meteor, a new metric that measures the accuracy of size and count. Third, current captioning datasets lack annotations of object size and count, limiting quantitative-aware training and evaluation, we therefore constructed size-annotated benchmarks by enriching existing datasets with explicit labels. Finally, experiments and ablation analysis verified the effectiveness of our framework and confirmed its superiority in generating quantitatively informative captions.

Although the proposed framework demonstrates strong performance, several limitations remain to be addressed in future work. The reliance on relative rather than absolute size measurements, which may limit applicability when sensor calibration data are required; 1) the restricted availability of large-scale remote-sensing datasets annotated with both object count and relative size; and 2) the computational overhead introduced by the graph construction and convolution processes.

Future work may focus on incorporating additional spatial semantics, such as elevation or object depth, and extending the framework to video-based remote sensing captioning tasks. In future practical applications, our proposed framework can assist in geo-observation tasks such as urban expansion monitoring, land-use change detection, and environmental assessment, where quantitatively descriptive captions can support interpretation and decision-making.

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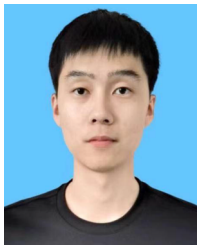
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